Sellekni · Business Model Canvas

SolveX 3.0 · Challenge 1 · Road emergencies in Algeria · A phone-based crash alert that reaches the emergency operations room by internet or plain SMS, with an estimated risk and a hospital suggestion.

Team: Cheikh Mohamed Tahar · Benkorich Abdenour · Ould Miloud Zakaria · Bouzid Mustapha · Baghdad Imad Eddine

31 Oct 2026 · prototype, synthetic data VERIFIED HYPOTHESIS PAID

The model in one sentence. The apps are free for everyone who sends an alert and, during the pilot, for the emergency services that receive it. Money comes from paid hardware (an OBD dongle and the Sellekni Box) for drivers and fleets who want stronger crash detection, then from institutional licences and insurers, with non-dilutive funding to start. Payers and prices are still to be validated.

8 · KEY PARTNERS

Who could help us?

- **Civil Protection** (14 / 1021) ops rooms: co-design, ground truth
- **Health ministry, SAMU, hospital EDs:** capability data, clinical review
- **Telecom operators:** SMS short code, coverage maps, LTE SIMs for the box
- **Emergency physicians, universities:** score calibration
- **Electronics makers and assemblers:** box prototype and production
- **ARPCE:** equipment homologation
- **DNSR:** statistics, road-safety campaigns
- **Incubators, startup-label committee**
- Later: **insurers, fleets** for distribution
- TO APPROACH · NONE AGREED

7 · KEY ACTIVITIES

What must we do?

- Build and maintain the app, gateway and dashboard
- Record real driving traces, tune the detector, measure false alarms
- Verify hospital capabilities; calibrate the score with emergency physicians
- **Hardware:** prototype the box, port the encoder to C, homologate
- Operator training, pilot measurement, legal (law 18-07 / 25-11)

6 · KEY RESOURCES

What do we need?

- **Built:** SLK1 protocol and rules engine (67 tests + Python cross-check), crash detector with replays (21 tests), dashboard, witness and hospital views in AR / FR / EN VERIFIED
- **Designed:** Sellekni Box (ESP32) and OBD link reference designs (see page 2)
- **To build:** verified hospital registry, hosting in Algeria, real driving data

2 · VALUE PROPOSITION

What problem do we solve?

- **Victim:** if you can't call, your phone alerts with your exact position. 30 s to say "I'm OK", then it sends itself. Works by plain SMS when data is weak; queues when there is no signal.
- **Witness:** report in 5 taps in the app, or 4 questions from a web link; no account.
- **14 operator:** one merged, located report with an estimated risk and its reasons; hospitals ranked by capability and travel time. The human decides; every action is logged.
- **Hospital ED:** a pre-alert before the patient arrives, acknowledged.
- **Hardware buyer:** the OBD dongle sharpens the speed drop; the Box's ±200 g sensor sees the real crash pulse that phones clip, with its own GNSS, LTE and backup power.
- **We don't** replace 14 / 1021, diagnose, or claim real-time bed data.

4 · CUSTOMER RELATIONSHIPS

How do we support them?

- Weekly co-design and a training day with operators; a named contact per partner
- Quiet app: no ads, no tracking outside an alert; SMS receipt to the sender
- "False alarm" feedback loop tunes the detector
- **Hardware:** install guide, firmware updates, warranty H

3 · CHANNELS

How do we reach them?

- Pilot: Android APK, witness web link, dashboard in any browser
- Later: Play Store (SMS-permission policy or composer fallback)
- Civil Protection days, driving schools, campuses, DNSR campaigns
- **Hardware:** fleet onboarding, insurer packs, car-accessory shops, online H

1 · CUSTOMER SEGMENTS

Who are they?

- **Free users:** drivers and passengers with Android phones (≈ 87 % of phones in Algeria) STATCOUNTER ; witnesses
- **Free receivers (pilot):** Civil Protection 14 / 1021, Gendarmerie 1055, Police 17, SAMU where present; hospital EDs
- **Hardware buyers:** drivers who want stronger detection; fleets and transport operators (buses, coaches, lorries) H
- **Institutional payers:** the State (B2G); motor insurers, ≈ 41.6 % of 2024 premiums TRÉSOR VIA ATLAS MAG.
- **Pilot early adopters:** one wilaya's ops room, one university hospital, one fleet

9 · COST STRUCTURE

What will it cost?

- **Fixed:** team time (the main cost); hosting in Algeria; map and routing hosting
- **Per incident:** about one SMS receipt and one hospital pre-alert, plus gateway SIMs
- **Hardware:** parts (ESP32, ±200 g sensor, LTE/GNSS modem, power), enclosure, box SIM data, homologation (Executive Decree 26-97), warranty
- **One-off:** medical and legal review, trademark (INAPI), test phones
- No supplier quotes yet, so no figures H

5 · REVENUE STREAMS

How will it be funded?

- **Apps: always free for citizens.** Nobody pays to be alerted.
- PAID **OBD dongle:** entry-level hardware sale to drivers and fleets
- PAID **Sellekni Box:** premium hardware sale, with an optional connectivity subscription for its SIM H
- **B2G:** licence and support contract for the ops-room software H
- **Fleets and insurers:** crash notification and data export subscription H
- **Now:** prizes, incubators, Algeria Startup Fund after labelling ROUTE VERIFIED
- No price set before interviewing 3 fleet managers and 2 insurers.

Sources: DNSR 2025 via Atlas Magazine; Direction Générale du Trésor via Atlas Magazine (insurance 2024); StatCounter (Android share); decrees 20-254 / 25-311 (startup label); Executive Decree 26-97 (homologation).

Hackathon prototype · not an official emergency service · no agreement with any partner today · in an emergency call 14 / 1021 · 1055 · 17

Sellekni · Hardware plan: OBD dongle and Sellekni Box

Page 2 of the Business Model Canvas · the paid tier behind block 5 (Revenue streams). Roadmap: designed, not built, not tested in a vehicle, not homologated.

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1 · PRODUCT TIERS

Free app, paid hardware, one packet

FREE · TODAY	FREE · TODAY	PAID · ROADMAP	PAID · ROADMAP
Phone app	**Witness**	**OBD dongle**	**Sellekni Box**
accelerometer, gyro, GPS; saturates on hard impacts	in the app or a web link, no account	wheel speed from the diagnostic port	±200 g sensor, own GNSS, LTE, backup power

every source → **same 160-character packet** internet first, SMS fallback → **same server** ops room + hospital → a human decides

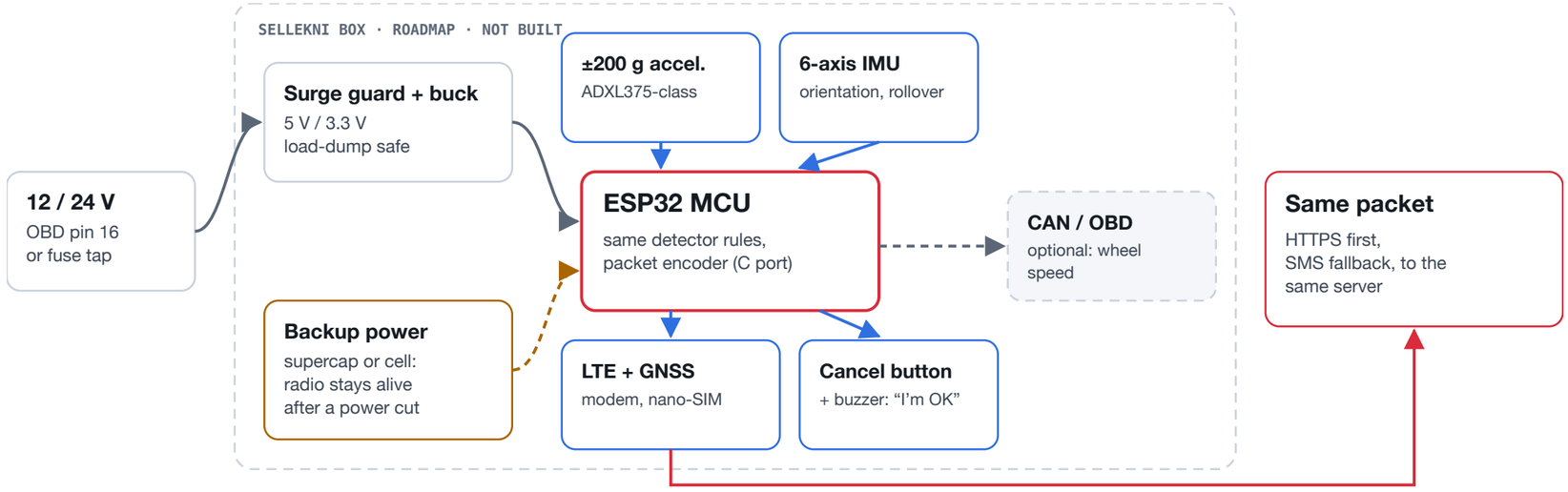
WHY HARDWARE

What a phone can’t see

- **Phones saturate** at about ±8–16 g, so a hard crash is clipped.
- **Sellekni Box:** a ±200 g sensor sees the real crash pulse; own GNSS, LTE and backup power, so it works even if the phone is thrown or broken.
- **OBD dongle:** the car’s own speed, faster than 1 Hz GPS.
- **Business:** the app stays free. Drivers and fleets buy the dongle (entry) or the box (premium, optional SIM subscription). No price set yet.

2 · SCHEMA · SELLEKNI BOX (ESP32)

Reference design and firmware flow



FIRMWARE FLOW · same rules as the phone

sample at max rate → ring buffer → trigger ≥ 4 g → gates: speed, time, stop → countdown + button → encode + send

- **Indicative parts** (none validated, no prices): ESP32-class module, ADXL375-class ±200 g accelerometer, 6-axis IMU, LTE modem with GNSS and antennas, nano-SIM holder, 12/24 V automotive buck with surge protection, supercap or small cell, optional CAN transceiver, enclosure.
- **Firmware:** the same detector rules as the phone; the packet encoder ported to C. Acceptance test: our 302 valid and 134 invalid test packets, byte for byte.
- **Network:** LTE, because Algeria plans to retire 2G/3G by 2029; check the module’s bands against the operators’ LTE bands before buying.
- **Before sale:** homologation under Executive Decree 26-97, GSMA TAC for the cellular module, automotive EMC tests, warranty and liability.

3 · SCHEMA · OBD LINK

What the car’s diagnostic port adds

Car	→	OBD dongle	→	Phone app	→	Same detector
OBD-II port under the dash pin 16: permanent +12 V		Bluetooth LE, ELM327-type initialised with AT commands		reads Mode 01 PID 0D vehicle speed, km/h		wheel speed + phone sensors sharper speed drop

It can:

- vehicle speed: Mode 01, PID 0D, in km/h
- engine speed: PID 0C
- faster speed readings than 1 Hz GPS

It can’t:

- read “airbag deployed”: no standard signal, it depends on the manufacturer
- B0001 is an airbag-circuit fault code, not a deployment

4 · PLAN

From phone to car, one stage at a time

NOW · FREE	NEXT · PAID	THEN · PAID	VISION
Phone	**OBD dongle**	**Sellekni Box**	**In the car**
impact, speed drop, stop; nothing to install	wheel speed over Bluetooth LE	±200 g sensor, GNSS, LTE, backup power	eCall-style link with makers and importers

- **We build hardware only when:** the phone pilot is stable; a teammate has the parts in hand; the firmware passes our test packets; it doesn’t delay the software.

Sources: Executive Decree 26-97 (equipment homologation); algerie-eco, 9 Apr 2026 (2G/3G retirement planned for 2029); SAE J1979 OBD-II PIDs 0C and 0D; full reference: docs/16 in our repository.